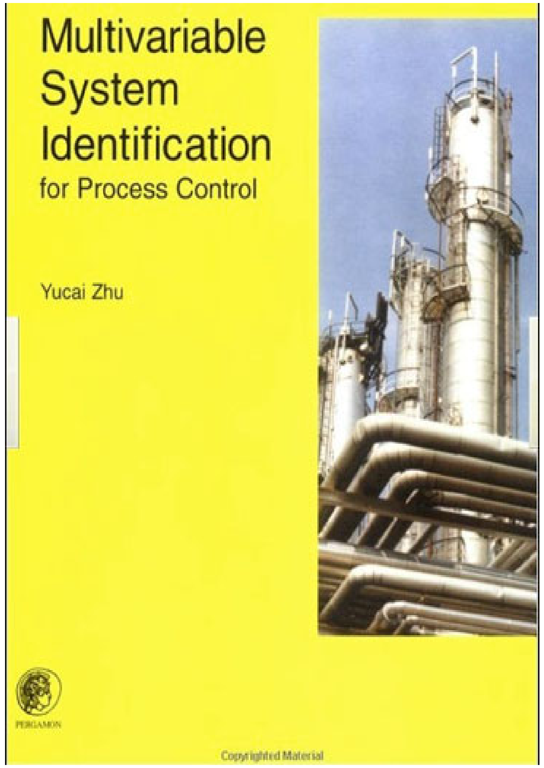
系统辨识：
动态数据建模的理论和方法

徐祖华教授
浙江大学控制科学与工程学院
工业智能与系统工程研究所

课程相关信息

- 教师
  - 徐祖华：工控所新楼413室
  - 电话： 18606523915
  - Email：zhxu@zju.edu.cn
  - http://mypage.zju.edu.cn/xuzh



- 参考资料

Zhu yucai. Multivariable system identification for process control. Elsevier,Lodon, 2001.

Chapter 1
Introduction to System Identification

System. A system is a collection of objects arranged in an ordered form to serve some purpose.

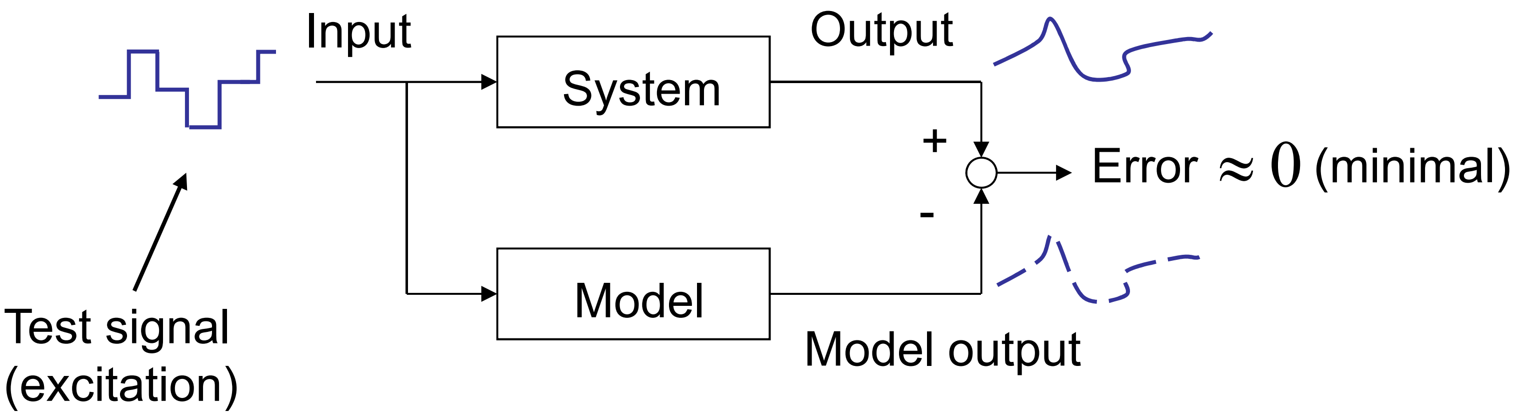
Process. A *process* is a processing plant that serves to manufacture homogeneous material or energy products in *process industries.* Process industries include: oil, chemicals, electrical power, paper, glass, mining, metals, cement, drugs, food.

Model. A model is a representation of the essential aspects of a system (process) which presents knowledge of that system (process) in a usable form (Eykhoff, 1974).

Mathematical models describe the relationships among the system variables in terms of differential and algebraic equations. The major part of the engineering field deals with the use of mathematical models for designs, simulations, forecasting and control/optimization.

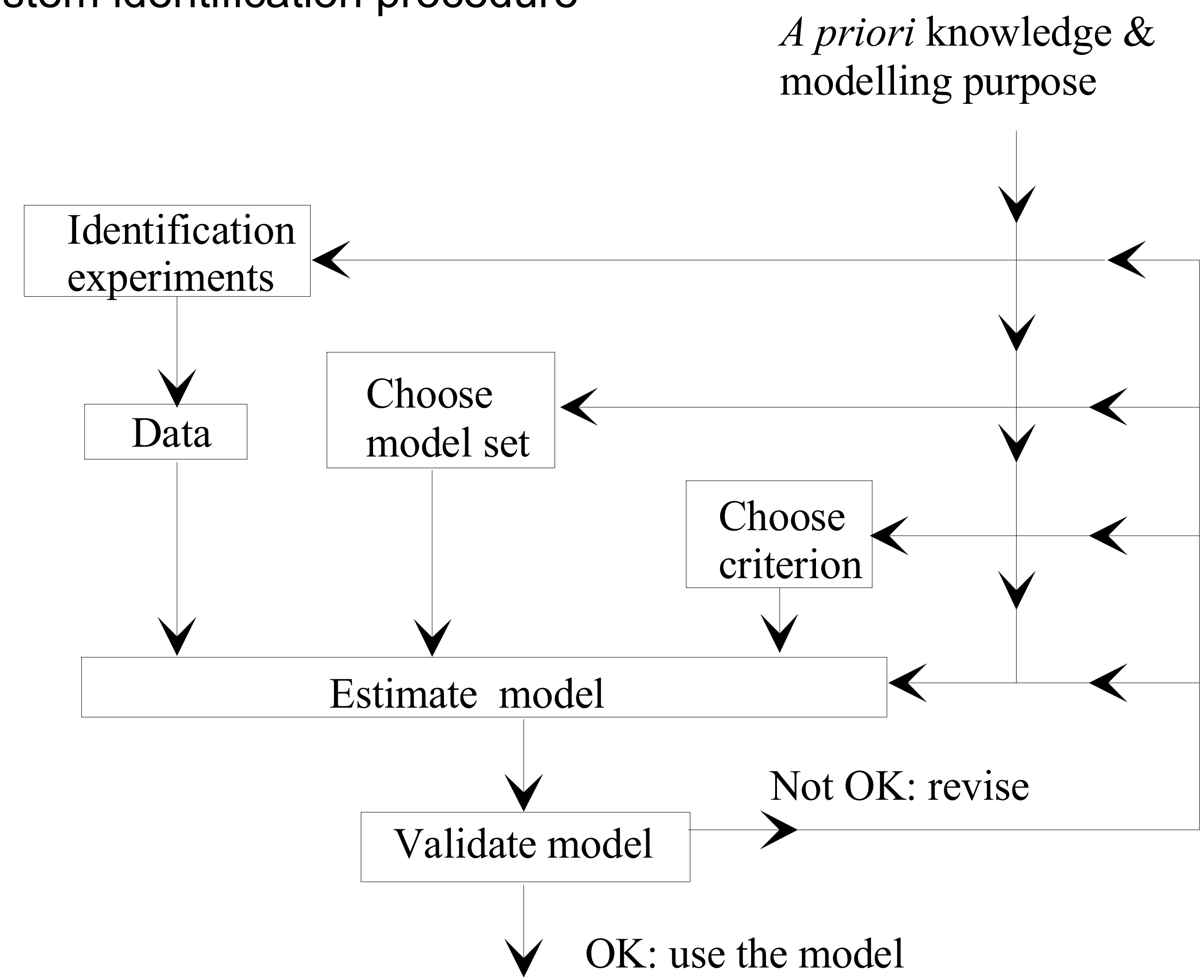
Introduction to System Identification

System Identification. System or process identification is the field of mathematical modelling of systems (processes) using **testing** data. In technical terms, *system identification* is defined by Zadeh (1962) as: *the determination on the basis of input and output, of a system (model) within a specified class systems (models), to which the system under* ***test*** *is equivalent (in terms of a criterion)*.



Introduction to System Identification

The system identification procedure



Introduction to System Identification

The system identification procedure has the following 4 steps:

1) Identification tests/experiments:
   To generate informative input-output data

2) Model order/structure selection:
   Linear or nonlinear, what structure (MIMO), what order?

3) Parameter estimation:
   To determine the model parameters using some optimization techniques.

4) Model validation:
   To check if the obtained model is good enough for its use (purpose) and, if not, give ways for the remedy.

Chapter 2
Models of Dynamic Systems and Signals

2.1 SISO Continuous-Time Models

A linear time-invariant SISO system

- Ordinary differential equation

$$y^{(n)}(t)+a_1 y^{(n-1)}(t)+\cdots+a_n y(t)=b_0 u^{(m)}(t)+b_1 u^{(m-1)}(t)+\cdots b_m u(t)$$

- Transfer function

$$G(s)=\frac{Y(s)}{U(s)}=\frac{b_0 s^m+b_1 s^{m-1}+\cdots+b_m}{s^n+a_1 s^{n-1}+\cdots+a_n}$$

2.1 SISO Continuous-Time Models

Model with delay d

$$y^{(n)}(t) + a_1 y^{(n-1)}(t) + \cdots + a_n y(t) = b_0 u^{(m)}(t-d) + b_1 u^{(m-1)}(t-d) + \cdots b_m u(t-d)$$

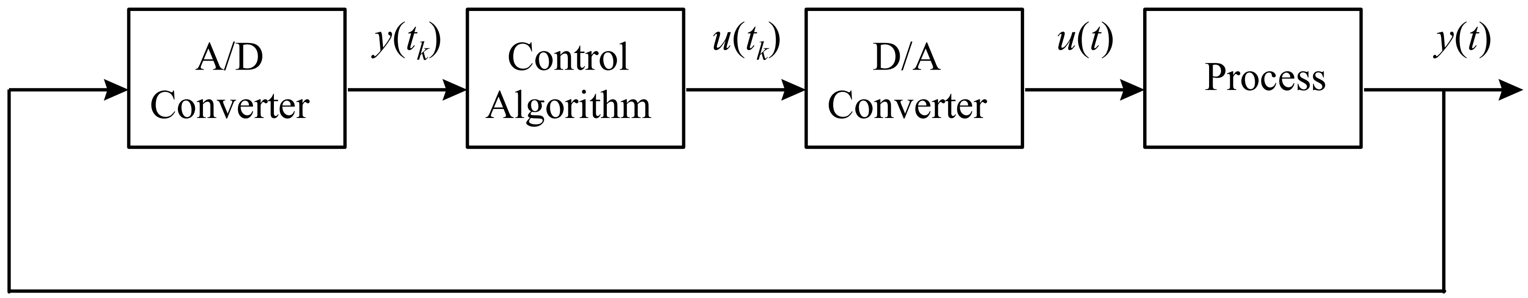
$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_m}{s^n + a_1 s^{n-1} + \cdots + a_n} \cdot e^{-sd}$$

Impulse response $g(\tau)$ and convolution

$$y(t) = \int_0^\infty g(\tau) u(t-\tau) d\tau$$

2.2 SISO Discrete-Time Models

Sampling of a computer control system using a zero-order hold



2.2 SISO Discrete-Time Models

Sampling an n-th order differential equation using zero-order hold:

$$y(t) + a_1 y(t-1) + \cdots a_n y(t-n) = b_1 u(t-1) + \cdots b_n u(t-n)$$

Transfer operator (with one delay)

$$G(q) = \frac{b_1 q^{-1} + \cdots + b_n q^{-n}}{1 + a_1 q^{-1} + \cdots + a_n q^{-n}}$$

With d delays (still finite order)

$$G(q) = \frac{b_1 q^{-1} + \cdots + b_n q^{-n}}{1 + a_1 q^{-1} + \cdots + a_n q^{-n}} \cdot q^{-d}$$

2.4 Models of Signals

■ Periodograms of Signals over Finite Intervals

For a finite sequence of signal $u(t), t = 1, 2, \cdots, N$

Discrete Fourier transform

$$U_N(\omega) = \frac{1}{\sqrt{N}} \sum_{t=1}^{N} u(t) e^{-i\omega t}$$

Inverse DFT

$$u(t) = \frac{1}{\sqrt{N}} \sum_{t=1}^{N} U_N(\frac{2\pi k}{N}) e^{-i2\pi kt/N}$$

Periodogram

$$\left|U_N(\omega)\right|^2$$

2.4 Models of Signals

Parsevals formula: The energy of the signal can be determined in both the time domain and the frequency domain

$$\sum_{k=1}^{N} \left| U_N\left(\frac{2\pi k}{N}\right) \right|^2 = \sum_{t=1}^{N} \left| u(t) \right|^2$$

Remark: This is the link between the time domain and the frequency domain

2.4 Models of Signals

■ Signal Spectra

Stochastic process {*v*(*t*)}: A set of signals which are, at each time instant, a random variable.

Stationary stochastic process: A stochastic process whose average properties do not change over time.

Let $\{v(t)\}$ be a stationary stochastic process

Autocorrelation function

$$R_v(\tau) = Ev(t)v(t-\tau)$$

Power spectrum

$$\Phi_v(\omega) = \sum_{t=-\infty}^{\infty} R_v(\tau)e^{-i\tau\omega}$$

2.4 Models of Signals

Given two stationary stochastic process $\{v(t)\}$ and $\{s(t)\}$, then

Cross-correlation function

$$R_{vw}(\tau) = Ev(t)w(t-\tau)$$

Cross-power spectrum

$$\Phi_{vw}(\omega) = \sum_{t=-\infty}^{\infty} R_{vw}(\tau)e^{-i\tau\omega}$$

If

$$s(t) = G(q)v(t)$$

Then

$$\Phi_s(\omega) = \left|G(e^{i\omega})\right|^2 \Phi_v(\omega)$$

$$\Phi_{sv}(\omega) = G(e^{i\omega})\Phi_v(\omega)$$

2.4 Models of Signals

White noise process $\{e(t)\}$: a sequence of independent and identically distributed random variables of zero mean and variance R.

$$R_e(\tau) = Ee(t)e(t-\tau) = \begin{cases} R & \tau = 0 \\ 0 & \tau \neq 0 \end{cases}$$

An important result: Every stationary stochastic process $\{v(t)\}$ can be generated by passing a white noise process through a stable minimum-phase filter (Astrom和Wittenmark, 1984)

$$v(t) = H(q)e(t)$$

2.5 Linear Process with Disturbances

$$\begin{aligned} y(t) &= G(q)u(t) + v(t) \\ &= G(q)u(t) + H(q)e(t) \end{aligned}$$

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